

STAT6160 DATA ANALYTICS FOR BUSINESS

Assignment 2 Questions

Number of questions: 2, Total marks: 25, Weight: 25%

Due time and date: 11:59 pm AEST, Sunday, end of Week 11

Instructions:

- Prepare your assignment in WORD, Latex, R Markdown, or any other appropriate software system for document preparation.
- Submit a copy in PDF format via Canvas.
- Assignments submitted by other means (e.g., email) or forms (e.g., scanned copy) will attract no marks.
- Late Submission Penalty: As detailed in the Course Outline.
- It is expected that R is used to assist with calculations and preparation of appropriate graphs. All relevant R scripts and output must be included with your assignment. However, raw computer output without explanatory text is not acceptable. Answers must be written in clear English sentences clearly linked to appropriate supporting computer output.
- The assessment requires you to apply concepts from Modules 1-4 to apply the correct analysis to the various scenarios/data sets and to write up the results of a statistical analysis.
- The use of generative AI is somewhat permitted for this assessment. See more details at <https://www.newcastle.edu.au/currentstudents/study-essentials/assessment-and-exams/assessments-and-assignments/artificial-intelligence-inassessment>. When using generative AI, you should use Microsoft Copilot, for which the university has a licence which protects our university's intellectual property.



For each sub-question with the **hand-and-pen logo**, respond using **your own understanding**. If Copilot contributes to your reasoning, you must cite it accordingly.



For each sub-question with the **RStudio logo**, include the **R code** and the corresponding **R output**. If you use Copilot to help with the code, you must cite it.



Copilot For each sub-question with the Copilot logo, provide:

1. Your **draft answer**, showing all working and reasoning.
2. The **output generated by Microsoft Copilot**, including the prompt you used.
3. A **final answer** that combines your draft and Copilot's output, along with your **assessment of Copilot's response quality**.

End of Instructions. The assignment questions begin with Page 2.

Question 1 (Total 15 Marks)

You are a supermarket chain's data analyst. The company recently launched a pilot digital loyalty program and wants to investigate its effectiveness.

To conduct the analysis, the company provided you with random samples of weekly average spending for 50 customers (25 existing customers and 25 new customers):

1. Before the digital loyalty program is launched, and
2. After the digital loyalty program has been launched.

The resulting dataset **LoyaltyProgram.csv** contains:

Variable	Description
CustomerID	Anonymised unique customer identifier
SpendingBefore	Weekly average spending amount (\$) <i>before</i> the program
SpendingAfter	Weekly average spending amount (\$) <i>after</i> the program
CustomerType	“Existing” or “New”

1. Using an appropriate statistical test, examine whether *post-program* spending differs between Existing and New customers.
 - a.  Discuss which statistical test is the most appropriate. **[1 Mark]**
 - b.  As the sample sizes are relatively small, use QQ plots to visually inspect whether SpendingBefore and SpendingAfter approximately follow Normal distributions. **[1 Mark]**
 - c. Conduct the statistical test using $\alpha = 0.01$ as the significance level. Be sure to state the hypotheses, test statistics, null distribution (e.g., t-distribution with appropriate degrees of freedom), p-value, and the conclusion. **[5 marks]**
2. Using an appropriate statistical test, determine whether the loyalty program has significantly changed the average spending for all customers.
 - a.  Discuss which statistical test is the most appropriate. **[1 Mark]**
 - b.   Conduct the statistical test using $\alpha = 0.01$ as the significance level. State the hypotheses, test statistics, null distribution (e.g., t-distribution with appropriate degrees of freedom), p-value, and the conclusion. **[5 marks]**

3.  **Copilot** Work with Copilot to double-check if you have chosen the correct type of statistical tests in (1) and (2). Provide the prompts you used in Copilot, along with Copilot's exact outputs. Then, evaluate how well Copilot performed. For example, whether Copilot has chosen the appropriate tests. If Copilot recommends a different test, explain why you accept or reject its advice. **[2 Marks]**

Question 2 (Total 10 Marks)

The file StoreSales.data provides data for company-owned stores across New South Wales. Each row represents a single observation: the sales performance of one store in one randomly selected month. For each region, stores were randomly selected, and their monthly sales data were independently and randomly collected.

Variable	Description
StoreID	Anonymised unique store identifier (e.g., "S043")
Region	Store location tag: "North coast", "South coast", "Sydney" or "Central coast"
MonthlySales	Net sales revenue for a random month (thousands of Australian dollars)
AdvertisingBudget	Store-level spending on local media for the same month is in thousands of Australian dollars

1. Using an appropriate statistical test that avoids the multiple comparisons problem, determine whether the average MonthlySales differs across the regions using $\alpha = 0.05$ as the significance level.

- (a)  **R Studio** State the hypotheses, test statistics, null distribution, p-value, and the conclusion. **[5 marks]**
- (b)  **R Studio** If the previous test shows a significant result, conduct post-hoc tests to determine which regions differ in MonthlySales. If you did not find a significant result in part (a), explain the purpose of post-hoc tests and why they are not appropriate. **[2 marks]**
- (c)  **R Studio** What are the assumptions of the hypothesis test in part (a)? List each one and check whether they are satisfied. **[3 marks]**